RIG-ETALE IMPLIES RIG-SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author environment labelled lemma-rig-etale-rig-smooth; the proof environment immediately following it is its own proof

Editorial note. The label lemma-rig-etale-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the definition of rig-smooth homomorphisms via annihilation of Ext^1 of the completed naive cotangent complex (definition-rig-smooth-homomorphism)
- the definition of rig-etale homomorphisms via vanishing of multiplication on the completed naive cotangent complex (definition-rig-etale-homomorphism)

Source reference. Original reference 0GAY: sources/stacks-restricted/pinned-original-source.tex, source lines 1794–1794. Prerequisite pointer invoked at 1977; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAI: sources/stacks-restricted/pinned-original-source.tex, source lines 831–831. Prerequisite pointer invoked at 1976; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at 1971; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and let I be an ideal. Let B be an object of (0AL4). If B is rig-étale over (A, I), then B is rig-smooth over (A, I).*

Proof. Immediate from Definitions 0GAI and 0GAY. □

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